CONTACT: Daniel Amewoalor amewoalor98@gmail.com

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In silico design and characterization of a novel multi-epitope mRNA vaccine against hepatocellular carcinoma targeting overexpressed CDC25C and AURKA

Daniel Amewoalor
Department of Biochemistry and Biotechnology, KNUST, Kumasi-Ghana

Introduction

Hepatocellular carcinoma (HCC) is the most common form of liver cancer. In Ghana, liver cancer is the second leading neoplasm condition. HCC is caused by viral insertions, chromosome translocation, gene amplification, etc[1].

Treatment options for HCC include systemic therapy, resection, transplantation, and local ablation. Resistance to chemotherapies used for treatment has been reported[1]. Immune checkpoint inhibitors result in adverse effects due to their immune-modulating activity.

CDC25C and AURKA play significant roles in cell division however their overexpression has been reported in poor prognosis and cancer progression[2].

Hypothesis

Tumor-associated antigens are capable of inducing specific immune responses. AURKA and CDC25C are overexpressed antigens in HCC and correlated positively with antigen-presenting cell infiltration in previous studies reported[2]. This study therefore seeks to design, evaluate, and characterize mRNA vaccine using TAAs.

Methods

1. Retrieval of Protein Sequence and Epitope Prediction — UniProt, IEDB
2. Epitope Assessment and Selection of Top Candidates — Vaxijen 2.0, AllerTop, ToxinPred 2.0, IFNepitope
3. In silico Vaccine Construction
4. Structural Predictions and Vaccine Refinement — Psipred, AlphaFold 2, GalaxyRefine
5. Physicochemical and Stereochemical Analysis — EXPasy, ProSA, Procheck
6. Docking Studies, Molecular Dynamics and Immune Simulations — Cluspro, iMODS, C-ImmSim

Results

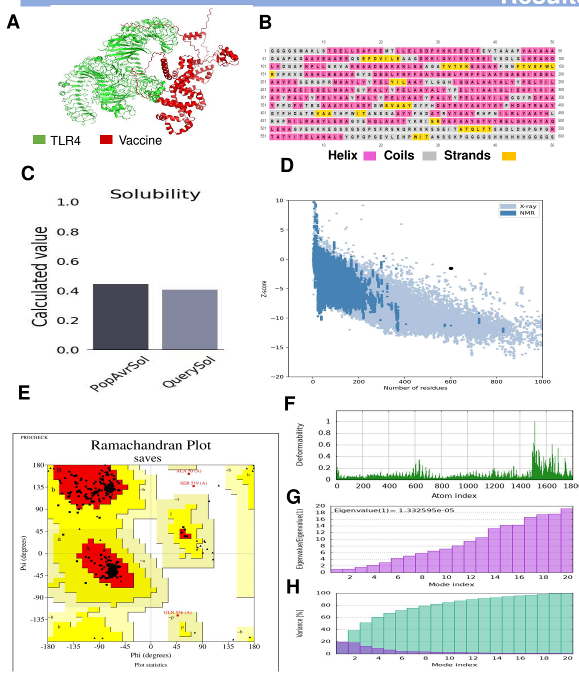

Figure 1: Structural and molecular dynamics analysis of the vaccine candidate. (A) 3D model of vaccine-TLR4 complex, (B) secondary structure, (C) solubility, (D) z-score plot, (E) Ramachandran plot, (F) deformability plot, (G) eigenvalue plot , and (H) variance plot

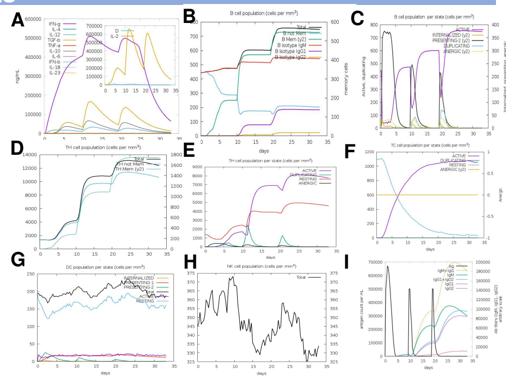

Figure 2: Simulation of the immune response to the vaccine construct. The vaccine was administered three times with a 28-day interval. The panels illustrate: (A)cytokines, (B, C) B-cells, (D, E) T_h cells, and (F) CD8 T-cells,(G)DC population, (H) NK cells, (H) Immunoglobulins

Conclusion

The candidate vaccine elicited a robust anti-tumor immune response during the simulation and has desirable physicochemical properties. The preliminary results indicate that the vaccine can be considered for further experimentation.

References

1. Llovet, J. M. et al. Hepatocellular carcinoma. Nat. Rev. Dis. Primer 7, 1–28 (2021).
2. Lu, T.-L., Li, C.-L., Gong, Y.-Q., Hou, F.-T. & Chen, C.-W. Identification of tumor antigens and immune subtypes of hepatocellular carcinoma for mRNA vaccine development. World J. Gastrointest. Oncol. 15, 1717–1738 (2023).